

EECS 492 – Homework 1

Written Solutions

Fall 2026

1 Designing Agents

1(a) PEAS description

The task environment contains a repeated sequence of tic-tac-toe games. At the end of a game, the board is cleared (or a fresh sheet is provided) and the next game begins. A PEAS description is therefore:

Component	Specification
Performance measure	Correctly recognize the board geometry, the symbols X and O , and empty cells; make only legal moves in empty cells; choose a move that is optimal for the agent's objective; correctly detect a win, loss, or tie; and reset and repeat the procedure reliably over arbitrarily many games. Secondary costs include drawing errors, sensing errors, unnecessary pen motion, and time.
Environment	A physical sheet of paper containing a 3×3 grid, the current markings, the pen and the board location, an opposing human or robot, and the sequence of past game outcomes. Between games the board is reset.
Actuators	A pen-moving mechanism, a pen-lowering/raising or contact mechanism, and a marker-drawing action that selects one of the nine cells. The agent may also signal the result of a game and request or perform a reset.
Sensors	A camera or other visual sensor for locating the grid, classifying X versus O , and detecting empty cells; position/contact feedback for the pen; and a state-change detector for recognizing the opponent's move and the terminal board state.

Assumptions. The paper remains within the sensing workspace, the grid is not intentionally destroyed, and illumination and handwriting vary within the sensor's operating range. The opponent is allowed to be either a person or another agent and may choose any legal move.

1(b) Environment characteristics

The classification below is for the physical robot described in the question, not for an idealized symbolic program. The distinction matters because sensing and pen motion introduce uncertainty even though the game rules themselves are simple.

Dimension	Classification	Justification
Observability	Partially observable	The true board state is not directly available as symbols. The agent must infer it from images, and glare, occlusion, ambiguous handwriting, or an imperfect classifier can hide whether a cell is empty or which marker it contains.
Number of agents	Multi-agent	The opponent selects actions that change the board and therefore changes the agent's future choices.
Transition model	Nondeterministic	The opponent's action is not controlled by the agent, and sensing or drawing can fail. If all perception and actuation were assumed perfect and the opponent's move were supplied as an action, the symbolic game transition would instead be deterministic.
Temporal dependence	Sequential	Each move changes the legal action set and can determine later wins, losses, or ties. The complete task is a sequence of such games, with a reset between episodes.
Change while deliberating	Dynamic	The opponent or the physical setup may change while the robot is sensing or planning. If the opponent is required to wait and the sheet is fixed during the robot's turn, the game is only semi-dynamic during that turn; the physical specification here permits changes, so "dynamic" is the conservative label.
State and actions	Discrete at the game level	There are finitely many cells, marker values, legal moves, and terminal results. Pen trajectories and camera measurements are continuous physical quantities, but they are abstracted into discrete game states and actions.
Model knowledge	Known at the rule level; uncertain at the physical level	The agent can be programmed with the tic-tac-toe rules and winning condition. It may not know the exact camera noise, pen-placement error, lighting, or opponent policy before calibration and experience.

2 K-Nearest Neighbors

The query point is

$$q = (8, 0, 3).$$

For Trial m , only the first m coordinates are used, so

$$d_m(q, x_i) = \left(\sum_{j=1}^m (q_j - x_{ij})^2 \right)^{1/2}.$$

2(a) Distances

The individual calculations are:

$$\begin{aligned}
\text{Trial 1: } d_1(q, x_1) &= |8 - 5| = 3, \\
d_1(q, x_2) &= |8 - 7| = 1, \\
d_1(q, x_3) &= |8 - 0| = 8, \\
d_1(q, x_4) &= |8 - 2| = 6, \\
d_1(q, x_5) &= |8 - 9| = 1, \\
d_1(q, x_6) &= |8 - 3| = 5.
\end{aligned}$$

For Trial 2, the first two coordinates give

$$\begin{aligned}
d_2(q, x_1) &= \sqrt{(8 - 5)^2 + (0 - 2)^2} = \sqrt{13} = 3.606, \\
d_2(q, x_2) &= \sqrt{(8 - 7)^2 + (0 - 6)^2} = \sqrt{37} = 6.083, \\
d_2(q, x_3) &= \sqrt{(8 - 0)^2 + (0 - 6)^2} = \sqrt{100} = 10.000, \\
d_2(q, x_4) &= \sqrt{(8 - 2)^2 + (0 - 7)^2} = \sqrt{85} = 9.220, \\
d_2(q, x_5) &= \sqrt{(8 - 9)^2 + (0 - 2)^2} = \sqrt{5} = 2.236, \\
d_2(q, x_6) &= \sqrt{(8 - 3)^2 + (0 - 5)^2} = \sqrt{50} = 7.071.
\end{aligned}$$

For Trial 3, all three coordinates give

$$\begin{aligned}
d_3(q, x_1) &= \sqrt{(8 - 5)^2 + (0 - 2)^2 + (3 - 8)^2} = \sqrt{38} = 6.164, \\
d_3(q, x_2) &= \sqrt{(8 - 7)^2 + (0 - 6)^2 + (3 - 1)^2} = \sqrt{41} = 6.403, \\
d_3(q, x_3) &= \sqrt{(8 - 0)^2 + (0 - 6)^2 + (3 - 0)^2} = \sqrt{109} = 10.440, \\
d_3(q, x_4) &= \sqrt{(8 - 2)^2 + (0 - 7)^2 + (3 - 4)^2} = \sqrt{86} = 9.274, \\
d_3(q, x_5) &= \sqrt{(8 - 9)^2 + (0 - 2)^2 + (3 - 4)^2} = \sqrt{6} = 2.449, \\
d_3(q, x_6) &= \sqrt{(8 - 3)^2 + (0 - 5)^2 + (3 - 8)^2} = \sqrt{75} = 8.660.
\end{aligned}$$

Thus the requested table is:

Point	Trial 1	Trial 2	Trial 3
x_1	3.000	3.606	6.164
x_2	1.000	6.083	6.403
x_3	8.000	10.000	10.440
x_4	6.000	9.220	9.274
x_5	1.000	2.236	2.449
x_6	5.000	7.071	8.660

2(b) $k = 3$ classification

The training labels are $y_1 = A$, $y_2 = B$, $y_3 = A$, $y_4 = B$, $y_5 = A$, and $y_6 = B$.

- **Trial 1.** The sorted distances are 1 for x_2 , 1 for x_5 , and 3 for x_1 (the first two are tied). The three labels are B, A, A , so the majority is A . The nearest-neighbor set is $\{x_2, x_5, x_1\}$, up to the order of the tied points.
- **Trial 2.** The three smallest distances are for x_5, x_1, x_2 , with labels A, A, B . The prediction is $\boxed{A}$.
- **Trial 3.** The three smallest distances are for x_5, x_1, x_2 , with labels A, A, B . The prediction is $\boxed{A}$.

Therefore, all three trials classify the query as A .

3 Decision Tree

For a label set S , use

$$H(S) = - \sum_{c \in \{\text{Yes}, \text{No}\}} p_c \log_2 p_c, \quad \text{IG}(S, A) = H(S) - \sum_{v \in \text{Values}(A)} \frac{|S_v|}{|S|} H(S_v).$$

The data contain 6 “Yes” labels and 4 “No” labels. Therefore,

$$\begin{aligned} H(Y) &= -\frac{6}{10} \log_2 \frac{6}{10} - \frac{4}{10} \log_2 \frac{4}{10} \\ &= 0.970950594 \approx 0.971. \end{aligned}$$

3(a) Information gain at the root

Outlook. The Sunny branch has (1 Yes, 3 No), the Overcast branch has (2 Yes, 0 No), and the Rain branch has (3 Yes, 1 No). Hence

$$\begin{aligned} H(\text{Sunny}) &= -\frac{1}{4} \log_2 \frac{1}{4} - \frac{3}{4} \log_2 \frac{3}{4} = 0.811278, \\ H(\text{Overcast}) &= 0, \\ H(\text{Rain}) &= -\frac{3}{4} \log_2 \frac{3}{4} - \frac{1}{4} \log_2 \frac{1}{4} = 0.811278, \\ \text{IG}(Y, \text{Outlook}) &= 0.970951 - \frac{4}{10}(0.811278) - \frac{2}{10}(0) - \frac{4}{10}(0.811278) \\ &= \boxed{0.322}. \end{aligned}$$

Humidity. The High branch has (2 Yes, 3 No), while the Normal branch has (4 Yes, 1 No). Thus

$$\begin{aligned} H(\text{High}) &= -\frac{2}{5} \log_2 \frac{2}{5} - \frac{3}{5} \log_2 \frac{3}{5} = 0.970951, \\ H(\text{Normal}) &= -\frac{4}{5} \log_2 \frac{4}{5} - \frac{1}{5} \log_2 \frac{1}{5} = 0.721928, \\ \text{IG}(Y, \text{Humidity}) &= 0.970951 - \frac{5}{10}(0.970951) - \frac{5}{10}(0.721928) \\ &= \boxed{0.125}. \end{aligned}$$

Temperature. The Hot branch has (1 Yes, 2 No) and the Mild branch has (2 Yes, 1 No), so both have entropy 0.918296. The Cool branch has (3 Yes, 1 No), so its entropy is 0.811278. Therefore,

$$\begin{aligned} \text{IG}(Y, \text{Temperature}) &= 0.970951 - \frac{3}{10}(0.918296) - \frac{3}{10}(0.918296) - \frac{4}{10}(0.811278) \\ &= \boxed{0.095}. \end{aligned}$$

Since $0.322 > 0.125 > 0.095$, the first splitting attribute is $\boxed{\text{Outlook}}$.

3(b) Force Humidity as the first split

(i) Entropy in the two branches

The branch entropies, already computed from their label counts, are

$$\boxed{H(Y \mid \text{Humidity} = \text{High}) = 0.971}, \quad \boxed{H(Y \mid \text{Humidity} = \text{Normal}) = 0.722}.$$

(ii) Information gain for Humidity

The weighted post-split entropy is

$$\frac{5}{10}(0.970951) + \frac{5}{10}(0.721928) = 0.846439,$$

so

$$\boxed{\text{IG}(Y, \text{Humidity}) = 0.970951 - 0.846439 = 0.125}.$$

(iii) Split each branch on Temperature

For the High-humidity branch, Temperature=Hot contains (1 Yes, 2 No) and Temperature=Mild contains (1 Yes, 1 No). Consequently,

$$\begin{aligned} \text{IG}(\text{High}, \text{Temperature}) &= 0.970951 - \frac{3}{5}(0.918296) - \frac{2}{5}(1.000000) \\ &= \boxed{0.020}. \end{aligned}$$

For the Normal-humidity branch, Temperature=Cool contains (3 Yes, 1 No) and Temperature=Mild contains one Yes label. The latter branch is pure, so its entropy is zero:

$$\begin{aligned} \text{IG}(\text{Normal}, \text{Temperature}) &= 0.721928 - \frac{4}{5}(0.811278) - \frac{1}{5}(0) \\ &= \boxed{0.073}. \end{aligned}$$

3(c) Two problems with decision trees

1. **Overfitting.** If the tree keeps splitting until leaves are nearly pure, it can memorize noise or rare training examples. Its training error may be near zero while its test error is large. Limiting depth, requiring a minimum leaf size, or pruning reduces this problem.
2. **Instability under small data changes.** Greedy splitting makes an early choice that can change when one training point is added, removed, or slightly perturbed; the changed root then changes the entire subtree. Bagging or random forests reduce this variance by averaging many trees.

4 Regression

The training points are (7, 6), (2, 2), and (4, 3). With $h_{\mathbf{w}}(x) = w_1x + w_0$, the sum-of-squared-errors objective is

$$L(w_0, w_1) = (6 - 7w_1 - w_0)^2 + (2 - 2w_1 - w_0)^2 + (3 - 4w_1 - w_0)^2.$$

4(a) Least-squares weights

Set the two partial derivatives to zero. Equivalently, the normal equations are

$$\begin{aligned} 3w_0 + 13w_1 &= 11, \\ 13w_0 + 69w_1 &= 58, \end{aligned}$$

where

$$\sum x_i = 13, \quad \sum y_i = 11, \quad \sum x_i^2 = 49 + 4 + 16 = 69, \quad \sum x_i y_i = 42 + 4 + 12 = 58.$$

The determinant is $3 \cdot 69 - 13^2 = 38 > 0$, so the quadratic objective has a unique minimizer. Eliminating w_0 gives

$$(207 - 169)w_1 = 174 - 143 \implies 38w_1 = 31 \implies w_1 = \frac{31}{38} = 0.815789.$$

Substitution into the first normal equation gives

$$3w_0 = 11 - 13 \left(\frac{31}{38} \right) = \frac{15}{38}, \quad w_0 = \frac{5}{38} = 0.131579.$$

Thus

$$h(x) = \frac{31}{38}x + \frac{5}{38}.$$

4(b) Predictions for all five points

Using the same weights, the five predictions are

$$\begin{aligned} \hat{y}_1 &= h(7) = \frac{222}{38} = 5.842105, \\ \hat{y}_2 &= h(2) = \frac{67}{38} = 1.763158, \\ \hat{y}_3 &= h(4) = \frac{129}{38} = 3.394737, \\ \hat{y}_4 &= h(3) = \frac{98}{38} = 2.578947, \\ \hat{y}_5 &= h(-1) = -\frac{26}{38} = -0.684211. \end{aligned}$$

Therefore,

$$(\hat{y}_1, \hat{y}_2, \hat{y}_3, \hat{y}_4, \hat{y}_5) = (5.842105, 1.763158, 3.394737, 2.578947, -0.684211).$$

4(c) Overall loss on all five points

The residuals $y_i - \hat{y}_i$ are

$$\frac{3}{19}, \quad \frac{9}{38}, \quad -\frac{15}{38}, \quad -\frac{11}{19}, \quad -\frac{6}{19}.$$

Therefore,

$$\begin{aligned} L_{1:5} &= \left(\frac{3}{19}\right)^2 + \left(\frac{9}{38}\right)^2 + \left(\frac{15}{38}\right)^2 + \left(\frac{11}{19}\right)^2 + \left(\frac{6}{19}\right)^2 \\ &= \frac{485}{722} = 0.671745. \end{aligned}$$

Thus the requested overall loss is $\boxed{0.671745}$.

4(d) R^2 statistic

For all five observed outputs,

$$\bar{y} = \frac{6 + 2 + 3 + 2 - 1}{5} = \frac{12}{5} = 2.4,$$

and the total sum of squares is

$$\begin{aligned} \text{SST} &= \sum_{i=1}^5 (y_i - \bar{y})^2 \\ &= (6 - 2.4)^2 + (2 - 2.4)^2 + (3 - 2.4)^2 + (2 - 2.4)^2 + (-1 - 2.4)^2 \\ &= 25.2. \end{aligned}$$

Using $\text{SSE} = 0.671745$ from part (c),

$$\begin{aligned} R^2 &= 1 - \frac{\text{SSE}}{\text{SST}} = 1 - \frac{0.671745}{25.2} \\ &= \boxed{0.973343} \approx 97.3\%. \end{aligned}$$

On this five-point evaluation set, the model explains approximately 97.3% of the variation in the outputs. This is a strong fit for this small dataset; it should not be interpreted as a guarantee of the same performance on a larger or different population.